$t = (20 \times 10^{-6}\ F) \times (300\ V) \div (10 \times 10^{-3}\ A) = 0.60\ s$

24 Ans: D

Since $B = \frac{\mu_0 I}{2\pi r}$

$F = F_{XY} \propto (I)(2I)/r = 2I^2 / r$

Considering forces acting on wire Z due to X and Y,

$F_{XZ} \propto (I)(4I)/2r = 4I^2 / 2r = F$ pointing towards X

$F_{YZ} \propto (2I)(4I)/(r) = 8I^2 / r = 4F$ pointing away from Y

Since F_{XZ} and F_{YZ} are in opposite direction, F_{net} on $z = 4F - F = 3F$

25 Ans: C